

SAMAYUKTAM

Scale Test Report

Generated: 2026-06-09 17:40:18 UTC

1. Executive Summary

Metric	Value
Machine	7fe613510f66 (colab-cpu)
CPU	AMD EPYC 7B13
OS / GLIBC	Ubuntu 22.04.5 LTS ldd
Memory	47.05 GB
Total runs	32
Successful	32
Failed	0
Binary	prebuilt (Docker ubuntu:22.04, libzstd static)

2. Results by Scale

100k (100,000 floats)

Pattern	Status	Sort Time	Throughput	Output Size	PCMP SHA-256 (first 24)
uniform	SUCCESS	25 ms	4.0 Mf/s	437.2 KB	aebbb5c7d959bf67c1d7ee20...
gaussian	SUCCESS	43 ms	2.3 Mf/s	445.0 KB	b62565ac7db27abf60b8f6bf...
sorted_asc	SUCCESS	26 ms	3.8 Mf/s	192.2 KB	62cfc45ec41866162e773900...
sorted_desc	SUCCESS	19 ms	5.3 Mf/s	192.6 KB	0e2ebe6793d04dac027bbdbe...
edge_nan	SUCCESS	32 ms	3.1 Mf/s	435.8 KB	19cd91987b184751beaca0c9...
all_zeros	SUCCESS	11 ms	9.1 Mf/s	167.0 B	a95569efd22b449691c2a627...
constant	SUCCESS	11 ms	9.1 Mf/s	167.0 B	e2f10fa6bf35e5918ffccb8...
spike	SUCCESS	103 ms	1.0 Mf/s	7.6 KB	2cc5519040b6318f337fda66...

1M (1,000,000 floats)

Pattern	Status	Sort Time	Throughput	Output Size	PCMP SHA-256 (first 24)
uniform	SUCCESS	173 ms	5.8 Mf/s	4.2 MB	533105159ba98107f9577e72...
gaussian	SUCCESS	274 ms	3.6 Mf/s	4.3 MB	ff58a007d97808f2bdf17f98...
sorted_asc	SUCCESS	127 ms	7.9 Mf/s	1.3 MB	ae45288ba8cd54bb3936f3f3...
sorted_desc	SUCCESS	135 ms	7.4 Mf/s	1.4 MB	6883ba8f865c39cdd27eb0e5...
edge_nan	SUCCESS	271 ms	3.7 Mf/s	4.2 MB	dc20e9a02bcb122760730ba6...
all_zeros	SUCCESS	67 ms	14.9 Mf/s	304.0 B	fbd35f393d66cedfc3223c63...
constant	SUCCESS	62 ms	16.1 Mf/s	304.0 B	b7d9cc0e5d16585fda7be209...
spike	SUCCESS	190 ms	5.3 Mf/s	72.6 KB	cf4a9f115f1323057e3de0e2...

10M (10,000,000 floats)

Pattern	Status	Sort Time	Throughput	Output Size	PCMP SHA-256 (first 24)
---------	--------	-----------	------------	-------------	-------------------------

uniform	SUCCESS	1.79 s	5.6 Mf/s	40.3 MB	4aa6a2a4224fa55a13be8f2d...
gaussian	SUCCESS	3.17 s	3.2 Mf/s	40.9 MB	c55c45ede72f86973a735ebb...
sorted_asc	SUCCESS	2.61 s	3.8 Mf/s	27.8 MB	86fff5e7662c9ab80fa66bc0...
sorted_desc	SUCCESS	2.65 s	3.8 Mf/s	27.8 MB	07e9800f6e3be87fbb586cb1...
edge_nan	SUCCESS	2.98 s	3.4 Mf/s	40.2 MB	b7e6b4d4772ed98d24bb3b97...
all_zeros	SUCCESS	486 ms	20.6 Mf/s	1.6 KB	8eff8865a91af0a363e483e6...
constant	SUCCESS	509 ms	19.6 Mf/s	1.6 KB	2b06e31ab2e457f4187868fd...
spike	SUCCESS	727 ms	13.8 Mf/s	711.1 KB	bb72b74a4eeae11ee963f4e3...

100M (100,000,000 floats)

Pattern	Status	Sort Time	Throughput	Output Size	PCMP SHA-256 (first 24)
uniform	SUCCESS	19.51 s	5.1 Mf/s	409.7 MB	e3b24c80a6f1ecc905a180e0...
gaussian	SUCCESS	35.49 s	2.8 Mf/s	422.5 MB	058d08580f404b81e65bc51a...
sorted_asc	SUCCESS	26.21 s	3.8 Mf/s	324.0 MB	32154c600083863cfbe11bdd...
sorted_desc	SUCCESS	26.00 s	3.8 Mf/s	324.1 MB	87232db070f25c7cc4f97f49...
edge_nan	SUCCESS	34.52 s	2.9 Mf/s	410.6 MB	8a475b1dc89042f4b8b715c1...
all_zeros	SUCCESS	4.87 s	20.5 Mf/s	15.0 KB	0d4e806550f56390e6e5227c...

1B (1,000,000,000 floats)

Pattern	Status	Sort Time	Throughput	Output Size	PCMP SHA-256 (first 24)
uniform	SUCCESS	236.04 s	4.2 Mf/s	3.8 GB	e113d0108c3b1df3ff24670d...
gaussian	SUCCESS	481.56 s	2.1 Mf/s	3.9 GB	790df5ea407e37dc839df277...

3. Appendix — Full PCMP SHA-256 Hashes

100k

Pattern	SHA-256
uniform	aebbb5c7d959bf67c1d7ee2095b083191670e75574c29534d07ccd4101f78dbb
gaussian	b62565ac7db27abf60b8f6bf00cf67236b6fd00b618300cbebcc394ec618849c
sorted_asc	62cfc45ec41866162e7739008c8a174eb57163e6b6af2c5687ba7a591f8e1e55
sorted_desc	0e2ebe6793d04dac027bbdbefe6f50ef687a0571ceef466d37816e220f14700b
edge_nan	19cd91987b184751beaca0c9798058d7c8c4a54d56abcb4596db7d83336ebcd7
all_zeros	a95569efd22b449691c2a627d4115d60020062b27eba5089e2fee7ff329c3b10
constant	e2f10fa6bf35e5918fffcbb8fcff07211795b08b7617eb8ce674b484ebf5961e
spike	2cc5519040b6318f337fda66ea0bf314af4fb0f87f37cac41bbd75fe3d098e74

1M

Pattern	SHA-256
uniform	533105159ba98107f9577e726a8b808d16f5cffb62ee69bbe78a3db79adeb0e5
gaussian	ff58a007d97808f2bdf17f98ce5b070102e239bb5875bdcc34909b1015b43a8f
sorted_asc	ae45288ba8cd54bb3936f3f3d61f2ed6de29912abd9094ce45d77ee515ae7eb7
sorted_desc	6883ba8f865c39cdd27eb0e520ed148e95a78cae7cfb04924b3173c36f0557da
edge_nan	dc20e9a02bcb122760730ba64733a38bb55cd534fb75629f35e2904119e8eb0c
all_zeros	fbdb35f393d66cedfc3223c63cfab26f2c5c587077f7bf1122b8265cbc05af170
constant	b7d9cc0e5d16585fda7be209a85d2b7bc70f88949da444ccc76a15ab540b282

spike	cf4a9f115f1323057e3de0e2ca651274650703a4858a44044004008149855c4e
-------	--

10M

Pattern	SHA-256
uniform	4aa6a2a4224fa55a13be8f2d8dcccda1c0cee9d4d8d2208ba4c2fff1a686b742
gaussian	c55c45ede72f86973a735ebbf95302fc7fd0186e899f170224eed27330aa431d
sorted_asc	86fff5e7662c9ab80fa66bc06c9996669eb906ce3d63dded6b41b43a9bd32e09
sorted_desc	07e9800f6e3be87fbb586cb11382392e4cd530da66756ffdb00c2f6393ddb5cb
edge_nan	b7e6b4d4772ed98d24bb3b972f24d6a5a8633fe59c7b66c29847cc0ba148ee48
all_zeros	8efff8865a91af0a363e483e612a480bd6a2bd26e24bce7b5ece23d1e4b1750d0
constant	2b06e31ab2e457f4187868fd95fddfbb3d4a51dff2cdbe9c2b9f1a244608c738
spike	bb72b74a4eeae11ee963f4e30ab84a0bb5285e1a6379d321cb3024869ae827a5

100M

Pattern	SHA-256
uniform	e3b24c80a6f1ecc905a180e09dd58623527f98fd27fcf0e37181a0e2853d861b
gaussian	058d08580f404b81e65bc51aac3d2bd51593a417353a21aea919d3cef8a73e0c
sorted_asc	32154c600083863cfbe11bddb43a616403d1c3646710802bac779e62808d7630
sorted_desc	87232db070f25c7cc4f97f497cad5615acce6e96d34d07050df9d6c70423e9e3
edge_nan	8a475b1dc89042f4b8b715c1bb87a887a1d253c070ac5728bab432acd9c74410
all_zeros	0d4e806550f56390e6e5227cc9e36ee2a593ae3fea2134c4c56a7b3c97cd19d4

1B

Pattern	SHA-256
uniform	e113d0108c3b1df3ff24670dd0c52c62f03d5484553306ab54e79805ec8f4503
gaussian	790df5ea407e37dc839df27738d9746f2795a00c2f8113505c4dc2e06a24644b